

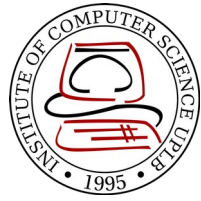
# *CMSC 137*

## *Data Communications and Networking*

Lecture Module  
06 - Performance

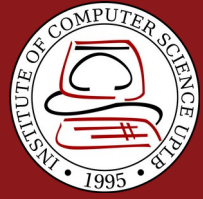


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# Module 6: Performance

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# Transmission Impairment

# Transmission Impairment



- Signals travel through transmission media which is not perfect
- What is sent is not what is received

Attenuation

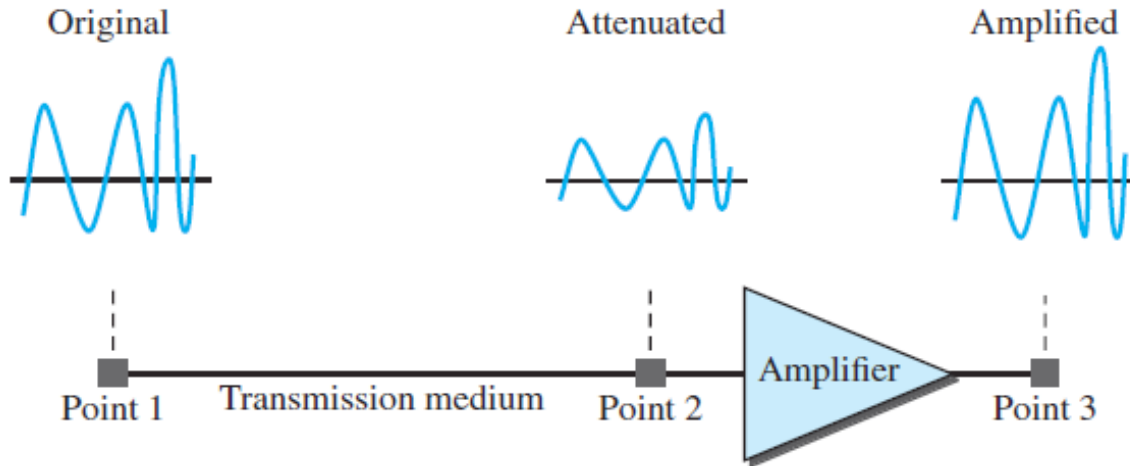
Distortion

Noise

# Attenuation



- Energy of a signal decreases as it travels through a medium
- **Amplifiers** are used to increase the signal

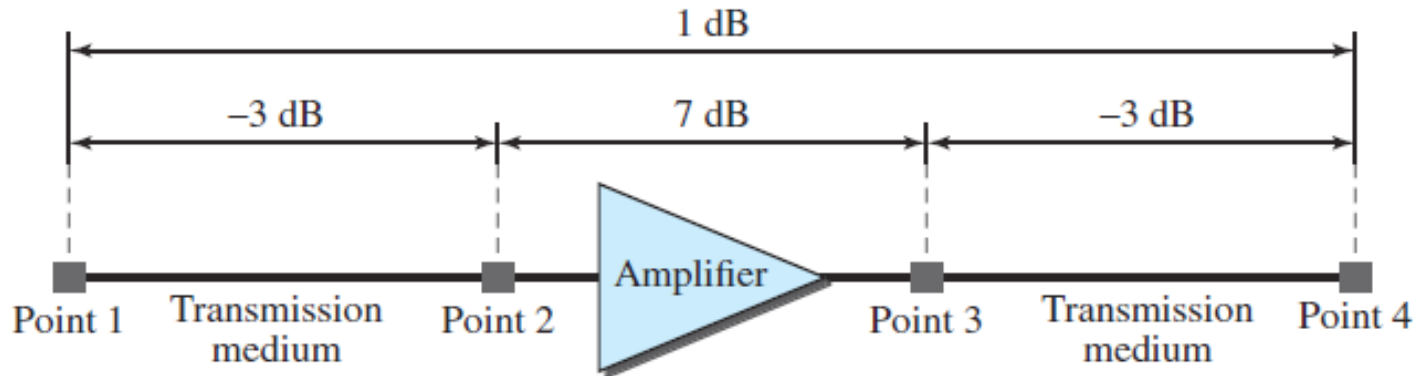


# Attenuation

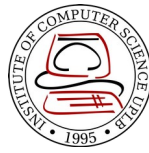


**Decibel** (dB) – measures the relative strengths of two signals or one signal at two different points

$$dB = 10 \log_{10} \frac{P_2}{P_1}$$



# Attenuation



*Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that  $P_2$  is  $(1/2)P_1$ . In this case, the attenuation (loss of power) can be calculated as*

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

*A loss of 3 dB (−3 dB) is equivalent to losing one-half the power.*

# Attenuation



*A signal travels through an amplifier, and its power is increased 10 times. This means that  $P_2 = 10P_1$ . In this case, the amplification (gain of power) can be calculated as*

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1}$$

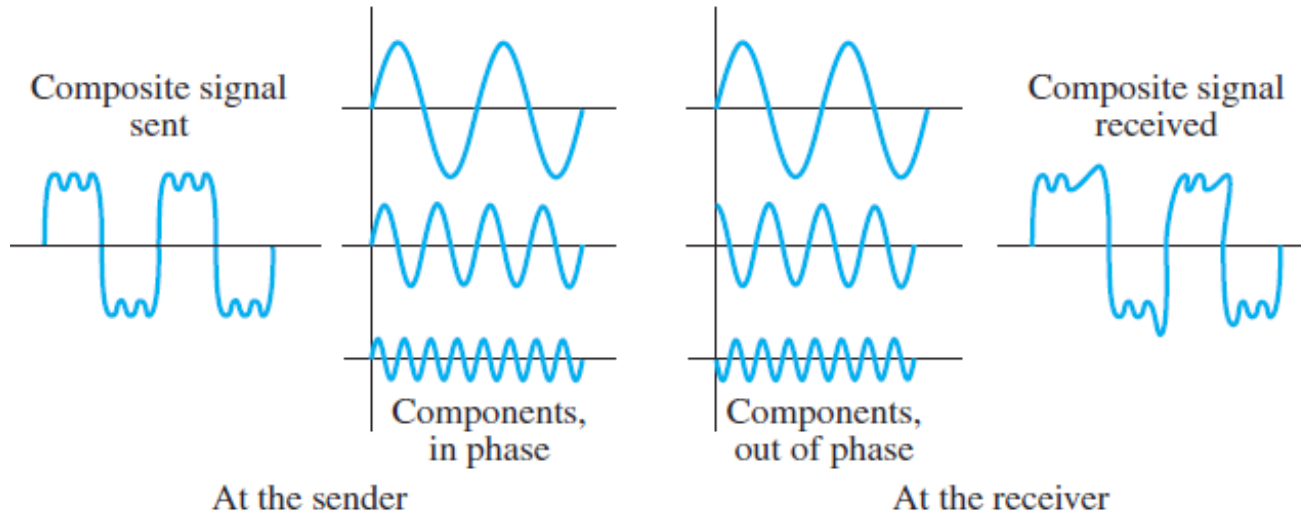
$$= 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$



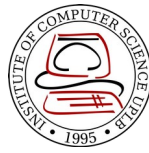
# Distortion



- Signal changes its form or shape
- Each signal component has its own propagation speed, components may be out of phase

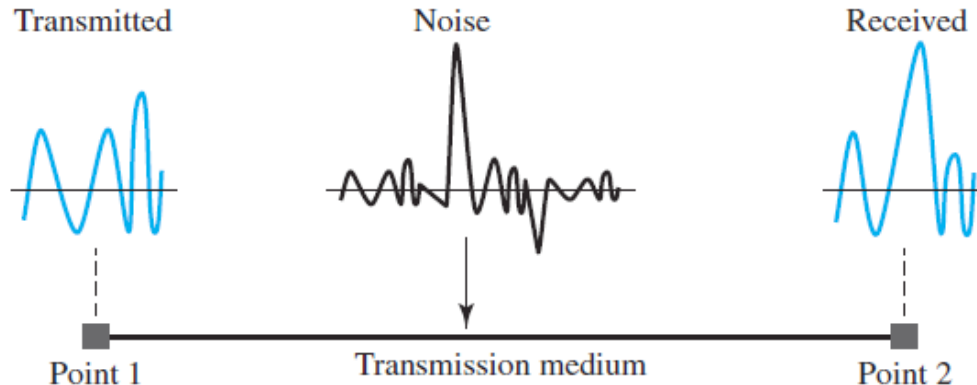


# Noise



## Types

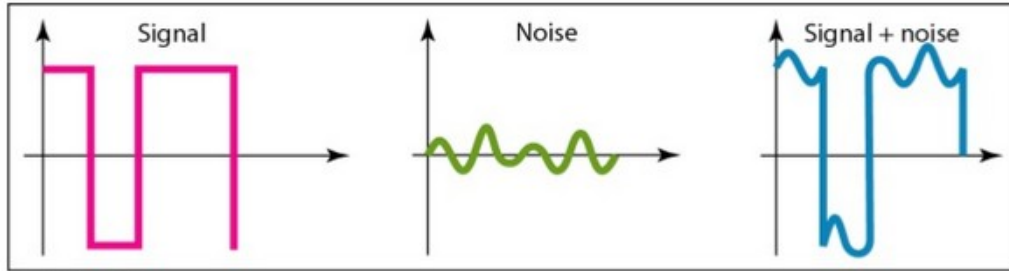
- **Thermal** – random motion of electrons
- **Induced** – comes from external sources (ex. appliances)
- **Crosstalk** – effect of one wire on the other
- **Impulse** – spike noise (ex. lightning, power lines)



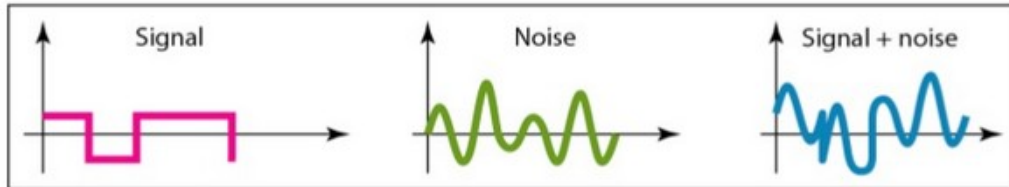
# Signal-to-Noise Ratio (SNR)



$$SNR = \frac{(\text{average signal power})}{(\text{average noise power})}$$

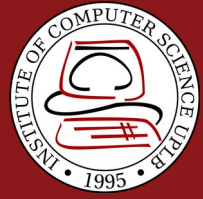


a. Large SNR



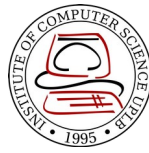
b. Small SNR

$$SNR_{dB} = 10 \log_{10} SNR$$



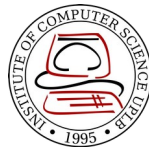
# Data Rate Limits

# Data Rate Limits



- How fast we can send data, in bits per second, over a channel:
  - Bandwidth available
  - Level of the signals we use
  - Quality of the channel (level of noise)
- Theoretical Formula:
  - Nyquist Bit Rate (Noiseless channel)
  - Shannon Capacity (Noisy channel)

# Data Rate Limits



- **Nyquist bit rate** formula defines the theoretical maximum bit rate (in bps)

$$Bit\ Rate_{Nyquist} = 2 \times bandwidth \times \log_2 L$$

- The Nyquist formula tells us *how many signal levels we need*
- Increasing the levels of a signal may reduce the reliability of the system

# Sample



What is the maximum bit rate of a noiseless channel with a bandwidth of 3 kHz transmitting two signal levels?

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

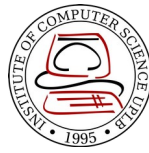
How many signal levels are needed to send at 265 kbps over a noiseless channel with a bandwidth of 20 kHz?

$$265,000 = 2 \times 20,000 \times \log_2 L$$

$$\log_2 L = 6.625$$

$$L = 2^{6.625} = 98.7 \text{ levels}$$

# Data Rate Limits



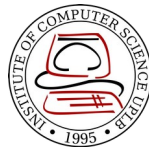
- **Shannon capacity** formula defines the theoretical highest data rate for a noisy channel (in bps)

$$Capacity = bandwidth \times \log_2(1 + SNR)$$

- The formula defines a characteristic of the channel, not the method of transmission
- The Shannon capacity gives us the upper limit



# Sample

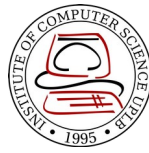


A telephone line has a typical bandwidth of 3000 Hz (300 to 3300 Hz) for data communications. SNR is 3162. What is the capacity?

$$C = 3000 \log_2(1 + 3162) = 3000 \log_2 3163$$

$$3000 \times 11.62 = 34,860 \text{ bps}$$

# Using both limits



In practice, both methods are used to find the limits and signal levels

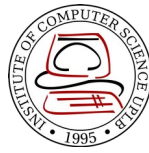
Example:

We have a channel with 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Use Shannon formula to find the upper limit

$$C = 10^6 \log_2(1+63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

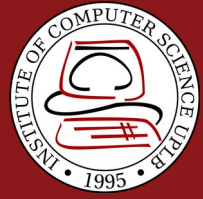
# Using both limits



The Shannon formula gives us 6 Mbps, the upper limit. For better performance, we choose something lower.

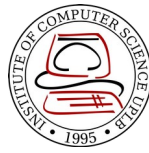
Let's say we use 4 Mbps. We use the Nyquist formula to find the number of signal levels

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L$$
$$L = 4$$



# Performance

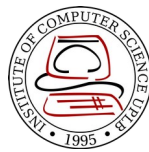
# Performance



- **Bandwidth**
  - Range of frequencies a channel can pass (Hz)
  - Number of bits per second that a channel can transmit (bps)
  - An increase in bandwidth in hertz means an increase in bandwidth in bits per second
- **Throughput** – how fast we can actually send data through a network
- **Latency** (Delay) – how long it takes for an entire message to completely arrive at the destination from the first bit

*Latency = propagation time + transmission time + queuing time + processing delay*

# Performance



- **Propagation Time** – time required for a bit to travel from source to the destination

$$\textit{Propagation time} = \frac{(\textit{Distance})}{(\textit{Propagation speed})}$$

- **Transmission Time** – time to transmit all the bits

$$\textit{Transmission time} = \frac{\textit{msize}}{\textit{bandwidth}}$$

- **Queuing Time** – time to hold the message before it is processed

# Performance

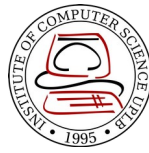


What are the propagation time and the transmission time for a 2.5 kB email if the bandwidth is 1 Gbps? Assume the distance between the sender and the receiver is 12,000 km and light travels at  $2.4 \times 10^8$  m/s

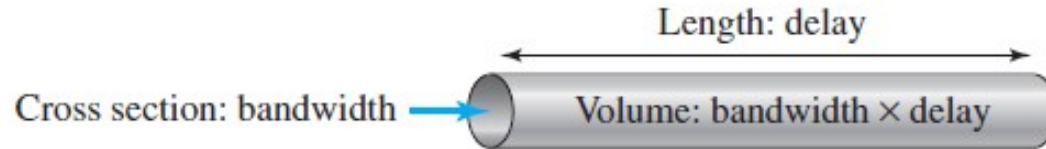
$$\textit{Propagation Time} = \frac{(12000 \times 1000)}{(2.4 \times 10^8)} = 50 \text{ ms}$$

$$\textit{Transmission Time} = \frac{(2500 \times 8)}{(10^9)} = 0.020 \text{ ms}$$

# Bandwidth-Delay Product

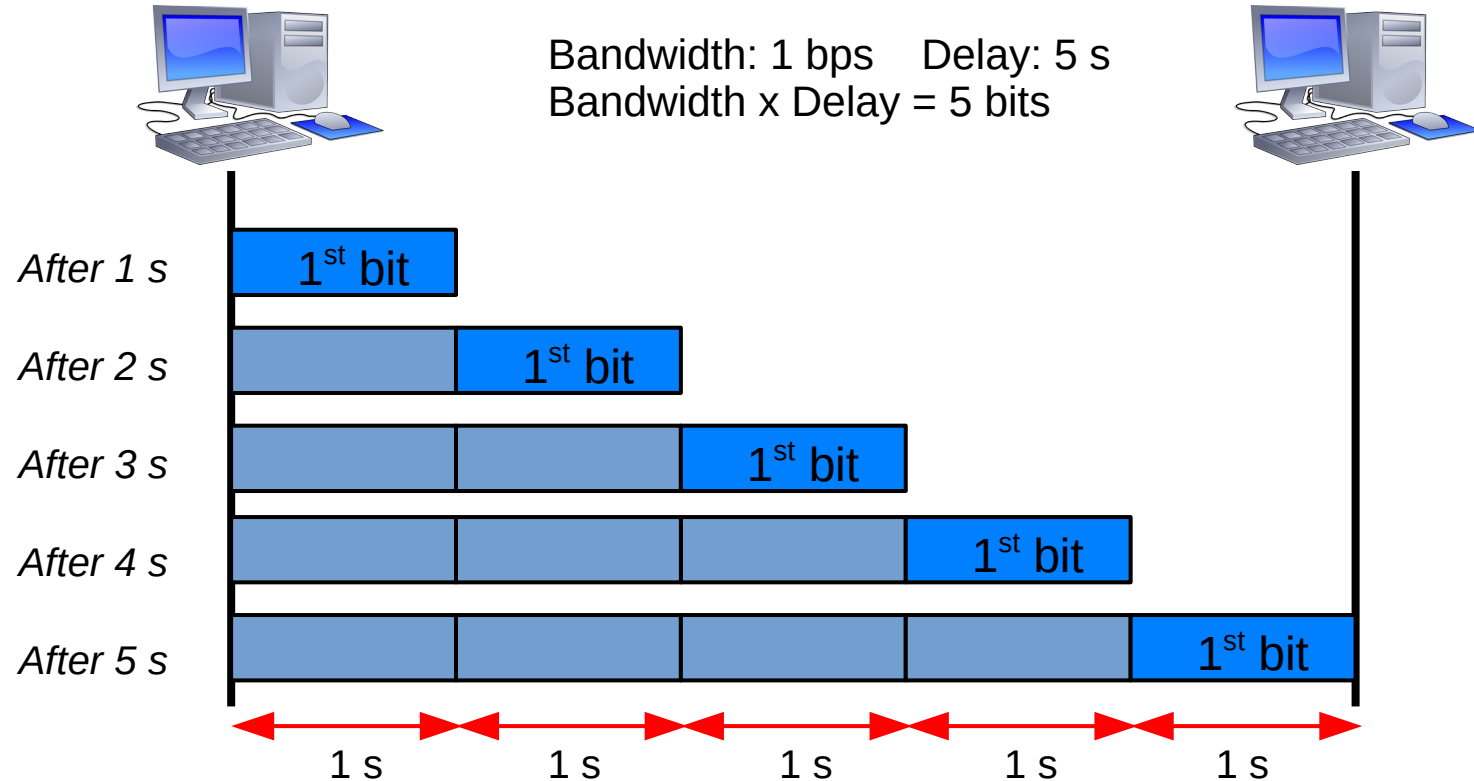
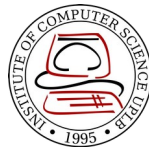


Defines the number of bits that can fill the link

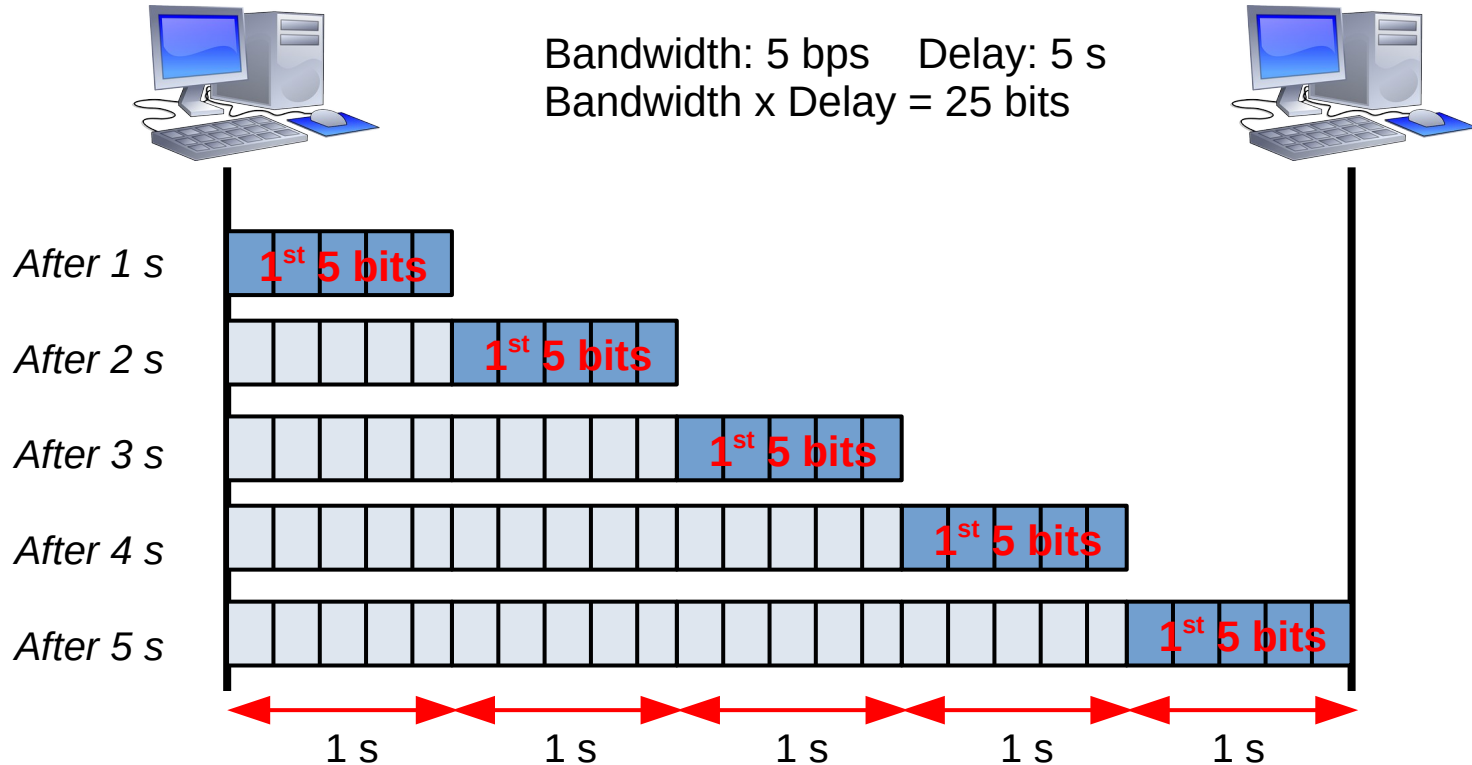
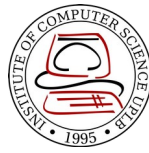




# Bandwidth-Delay Product



# Bandwidth-Delay Product



# Performance



- Jitter
  - If different packets of data encounter different delays and the application using the data at the receiver is time-sensitive

# References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.